Detection of GW, matched filtering

Search for GW

strain data $d(t)$: instrumental noise $n(t)$ + astrophysical GW signal $h(t)$: “数据分析基本定理： d = n + h”

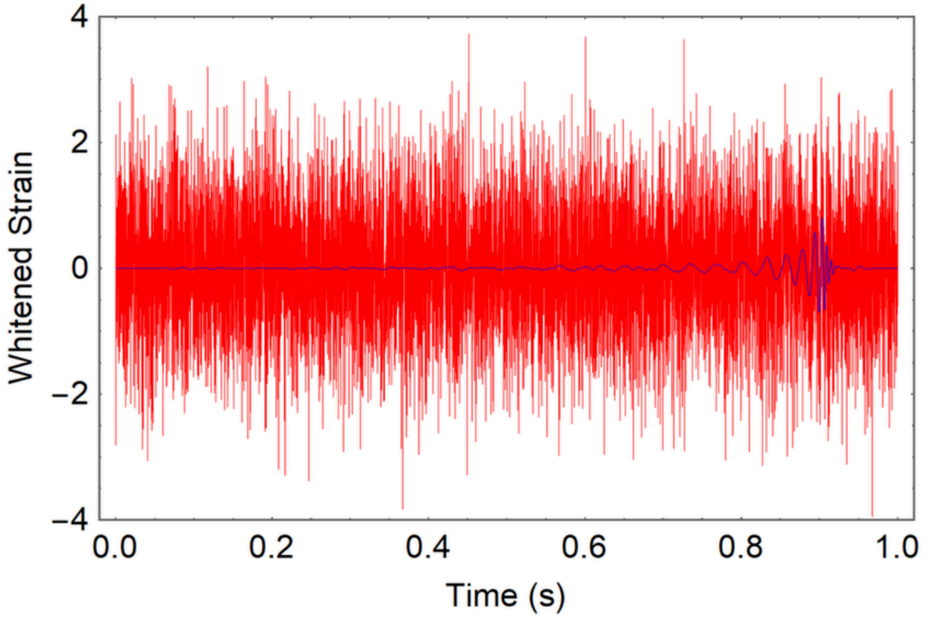
We want to answer:

- Is there a **signal** in the data?
- What are the **source parameters** of the signal?

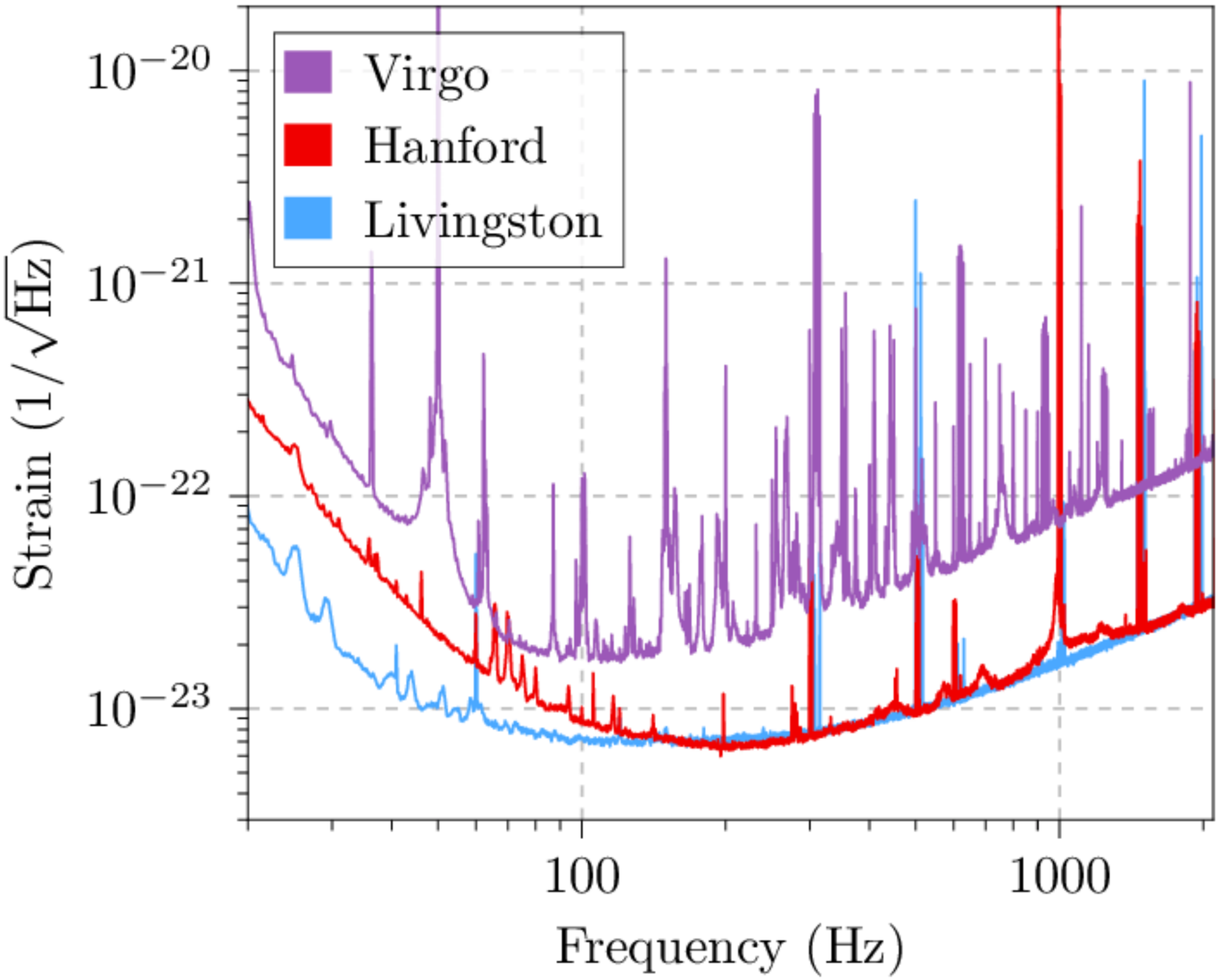
To answer, we need:

- Model for the **signal**
- Model for the **noise**

Noise is stochastic and therefore we need a **probabilistic model**

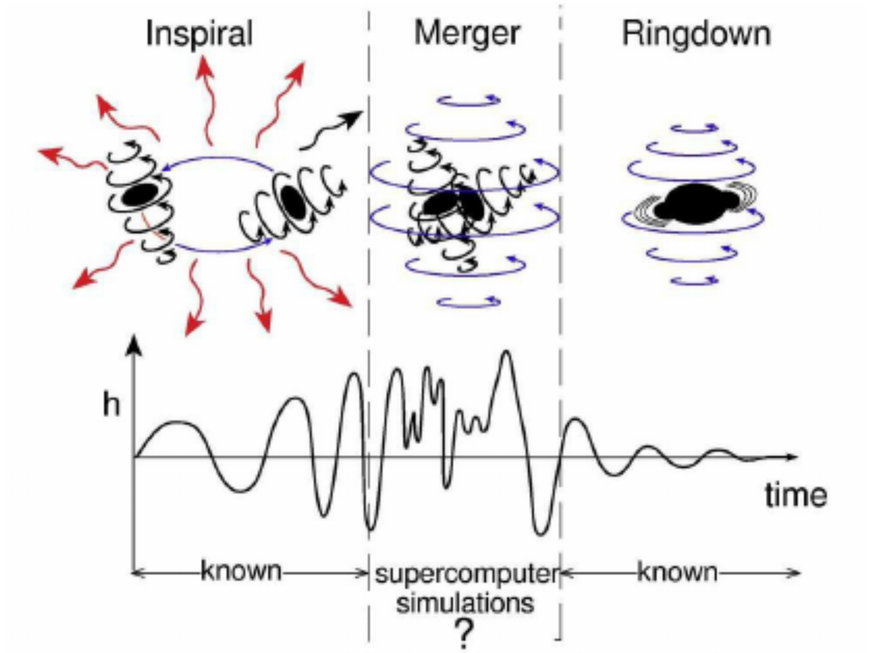


Statistical Modeling of Noise

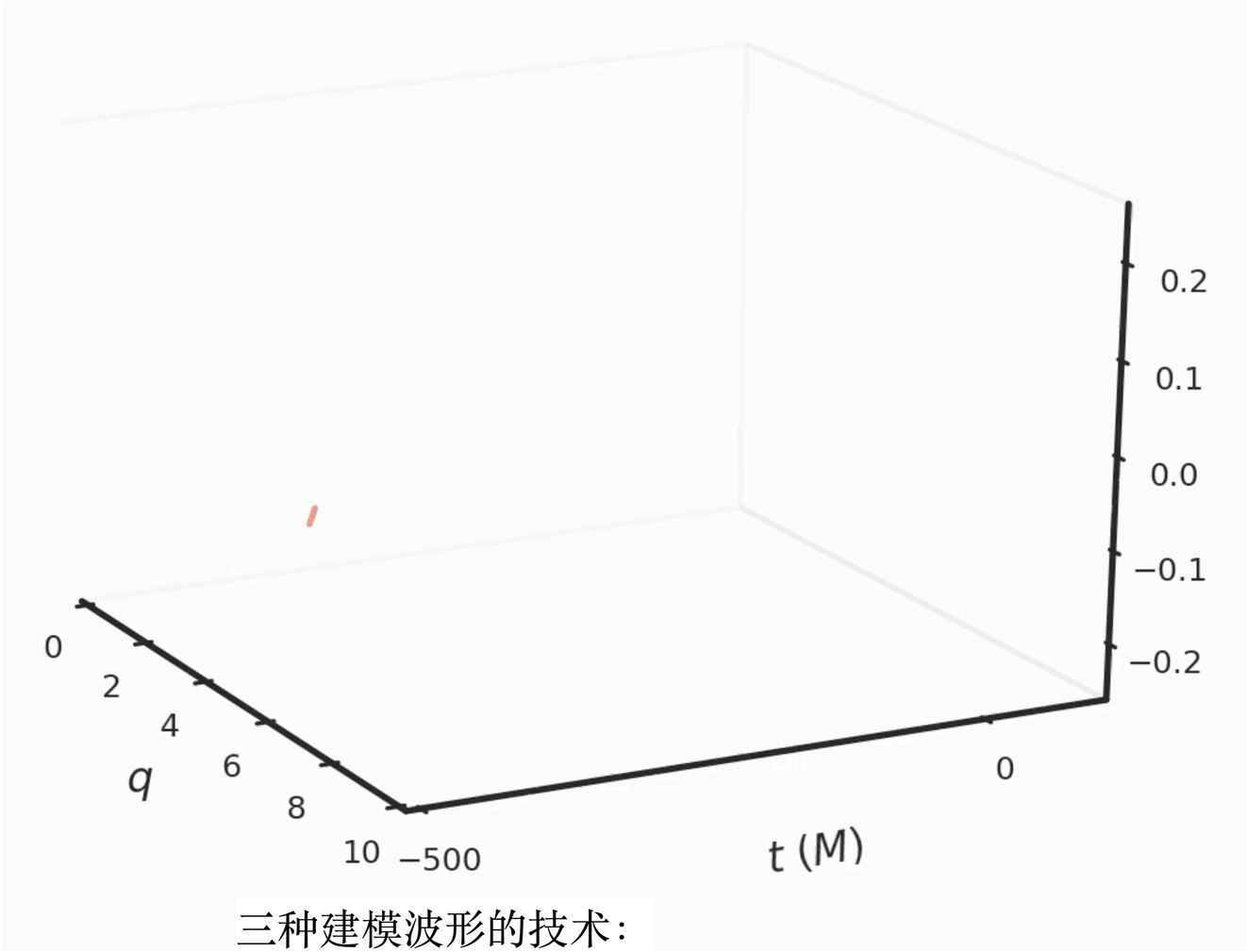


Power Spectral Density: 噪音功率谱密度

Modeling of Waveform



Kip Thorne's Sketch before 2005



三种建模波形的技术：

- 有效单体
- 纯现象学模板
- 数值相对论代理（Surrogate) 模型

New Waveform Model: SEOBNRv5EHM

SXS:BBH:1374

- Effective One Body (EOB) Hamiltonian:

$$H_{\rm EOB} = M\sqrt{1 + 2\nu\left(\frac{H_{\rm eff}}{\mu} - 1\right)}$$

- Together with the radiation reaction force $\mathscr{F}$, one has the binary dynamics:

$$\dot{r} = \frac{\partial H_{\rm EOB}}{\partial p_r} \qquad \dot{p}_r = -\frac{\partial H_{\rm EOB}}{\partial r} + \mathscr{F}_r$$

$$\Omega = \frac{\partial H_{\rm EOB}}{\partial p_\phi} \qquad \dot{p}_\phi = -\frac{\partial H_{\rm EOB}}{\partial \phi} + \mathscr{F}_\phi$$

- SEOBNRv5EHM: a new time-domain, inspiral-merger-ringdown multipolar waveform model for aligned spin, eccentric binary black holes

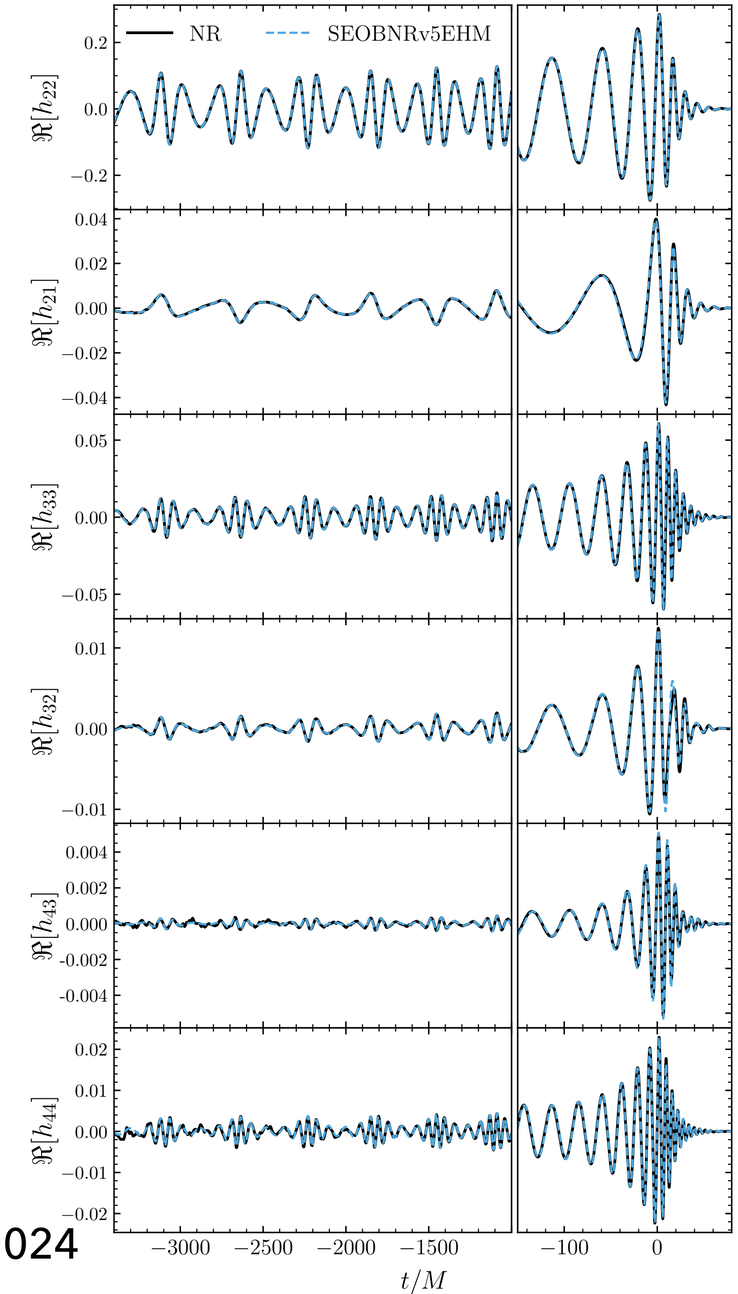
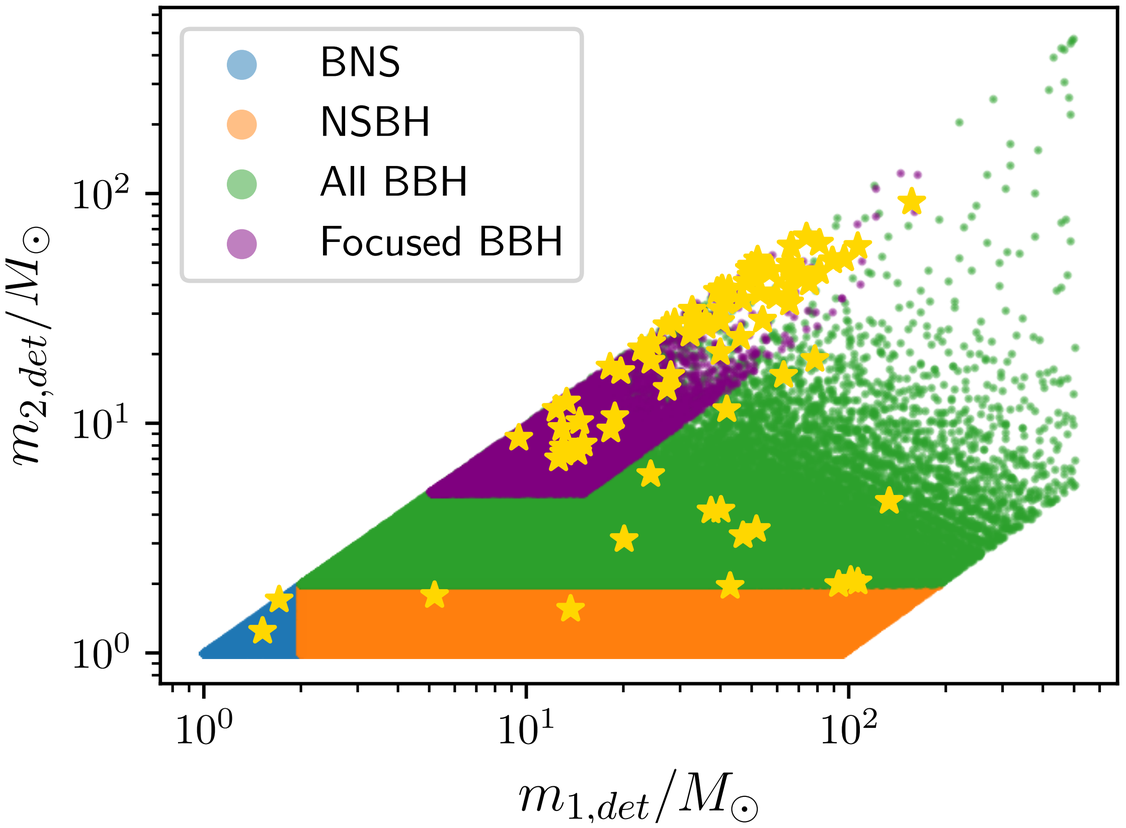
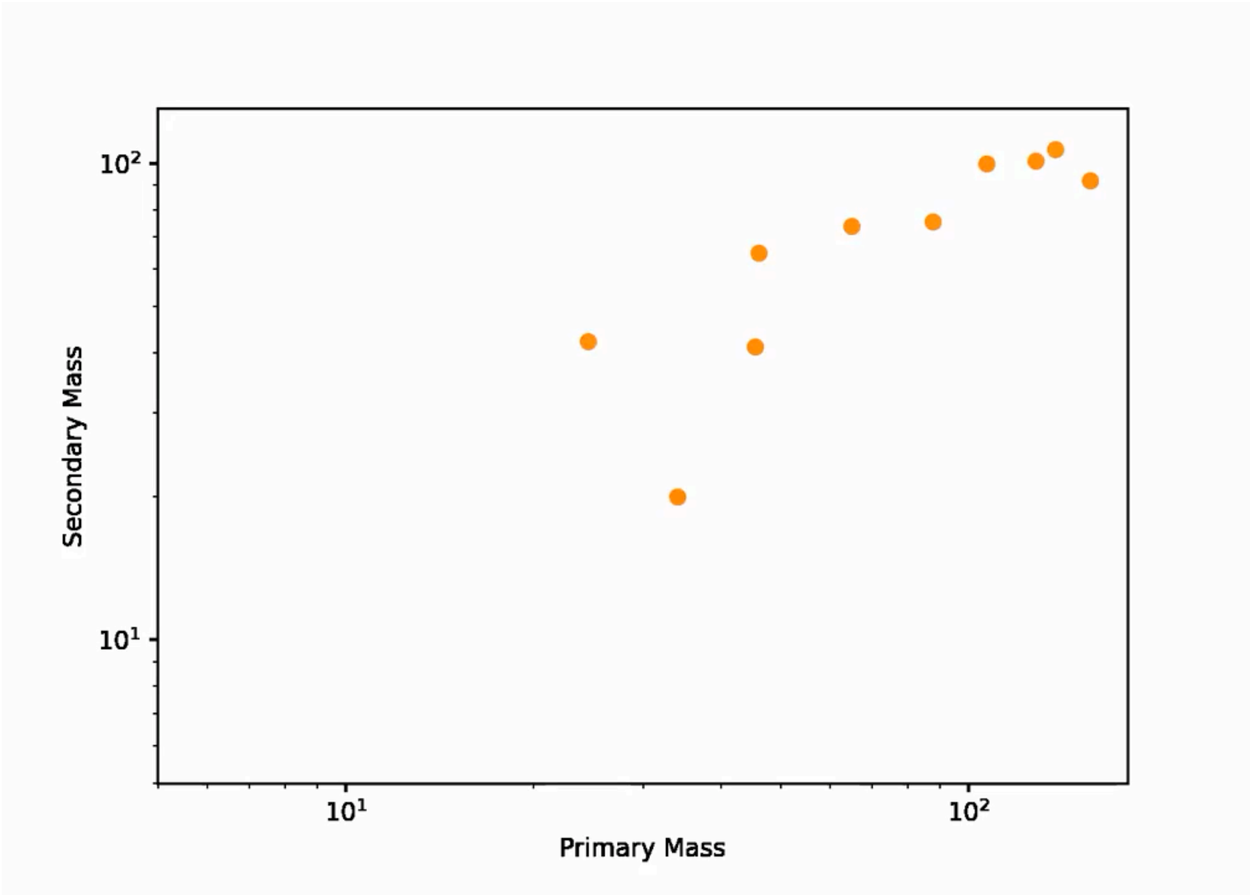


Image credit: Aldo Gamboa et al. 2024

Template Bank





如何又快又好地铺满整个空间？寻找引力波波形模版的低维表示

Hypothesis Testing

What is the probability of having a given GW signal $h(t)$ in the detector data?

- Assume we **know the noise** (we assume it is **gaussian, stationary** and therefore we can fully characterise it with its variance σ_n^2, that we can compute)
- Assume we **know the signal** $h(t)$ we are looking for

Null hypothesis: $\mathscr{H}_0$: $d(t) = n(t)$ no signal in the detector data

Signal hypothesis: $\mathscr{H}_1$: $d(t) = n(t) + h(t)$ signal in the detector data

信噪比与排序统计量(ranking statistic)

- 搜寻引力波可以归结为一个假设检验问题：
  - H_0: d = n, 其中$n \sim \mathcal{N}(0,\Sigma)$
  - H_1: d = n + h
- 两类错误：
  - 第一类错误, False Positive, P(接受$H_1|H_0$)
  - 第二类错误, False Negative, P(接受 $H_0|$ H_1)
- 根据Neyman-Pearson引理，对于简单假设H_0和H_1, 一个一致最优检验(Uniformly Most Powerful, UMP)是似然函数之比
  - $\ln \mathscr{L}_0 = -\frac{1}{2}\frac{d^2}{\sigma^2}$（示意性的写法）
  - $\ln \mathscr{L}_1 = -\frac{1}{2}\frac{(d-h)^2}{\sigma^2}$
- Match filter detection statistic = $\ln \mathscr{L}_1 - \ln \mathscr{L}_0 = \frac{dh}{\sigma^2} + const$

Match filtering in a Gaussian world

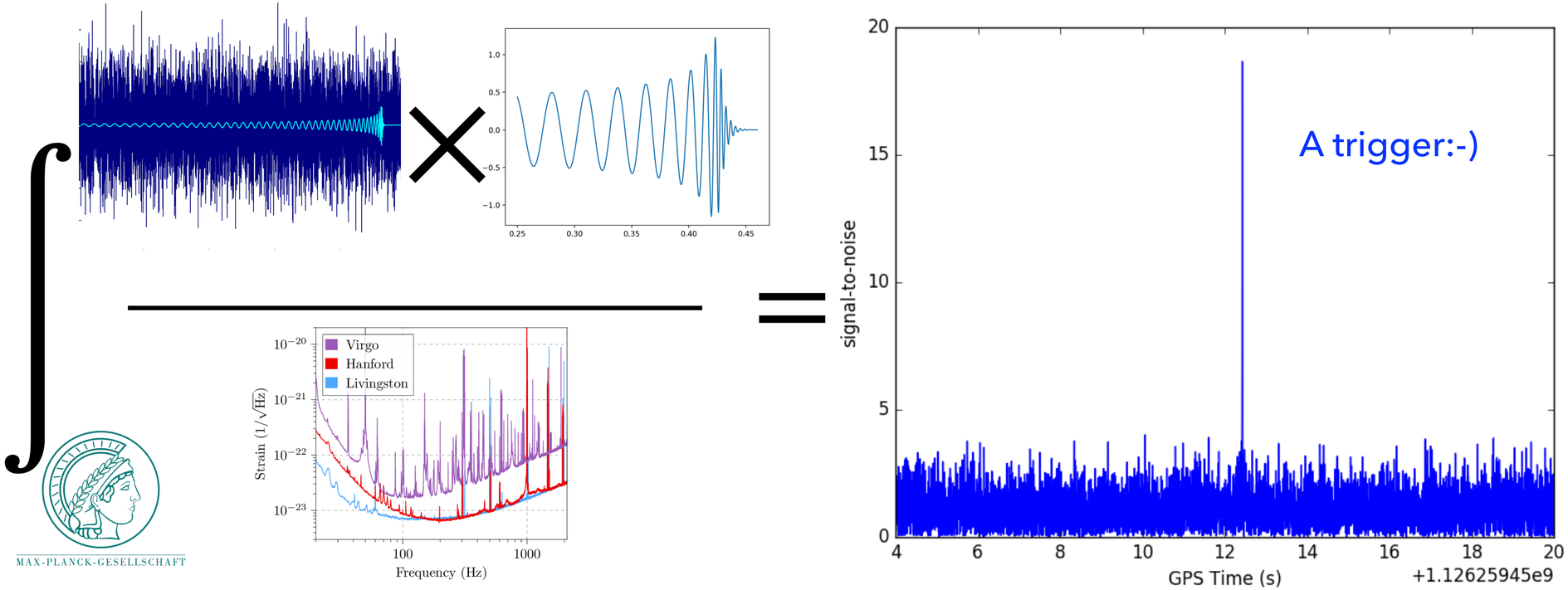
- Searching for CBC by match filtering (correlating between data and model)

Signal

Template

$$\frac{S}{N}(t) = 4\Re \int_0^\infty \frac{\tilde{s}(f)\tilde{h}^*(f)}{S_n(f)} \exp^{2\pi i f t}\, df$$

Noise power spectrum

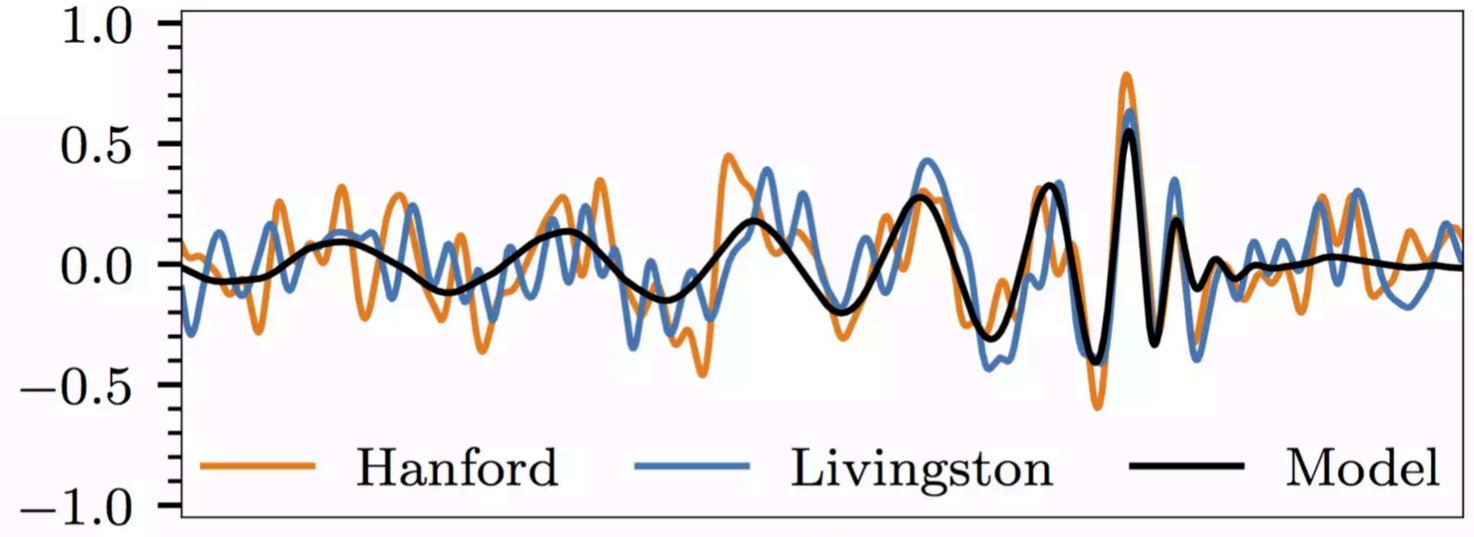


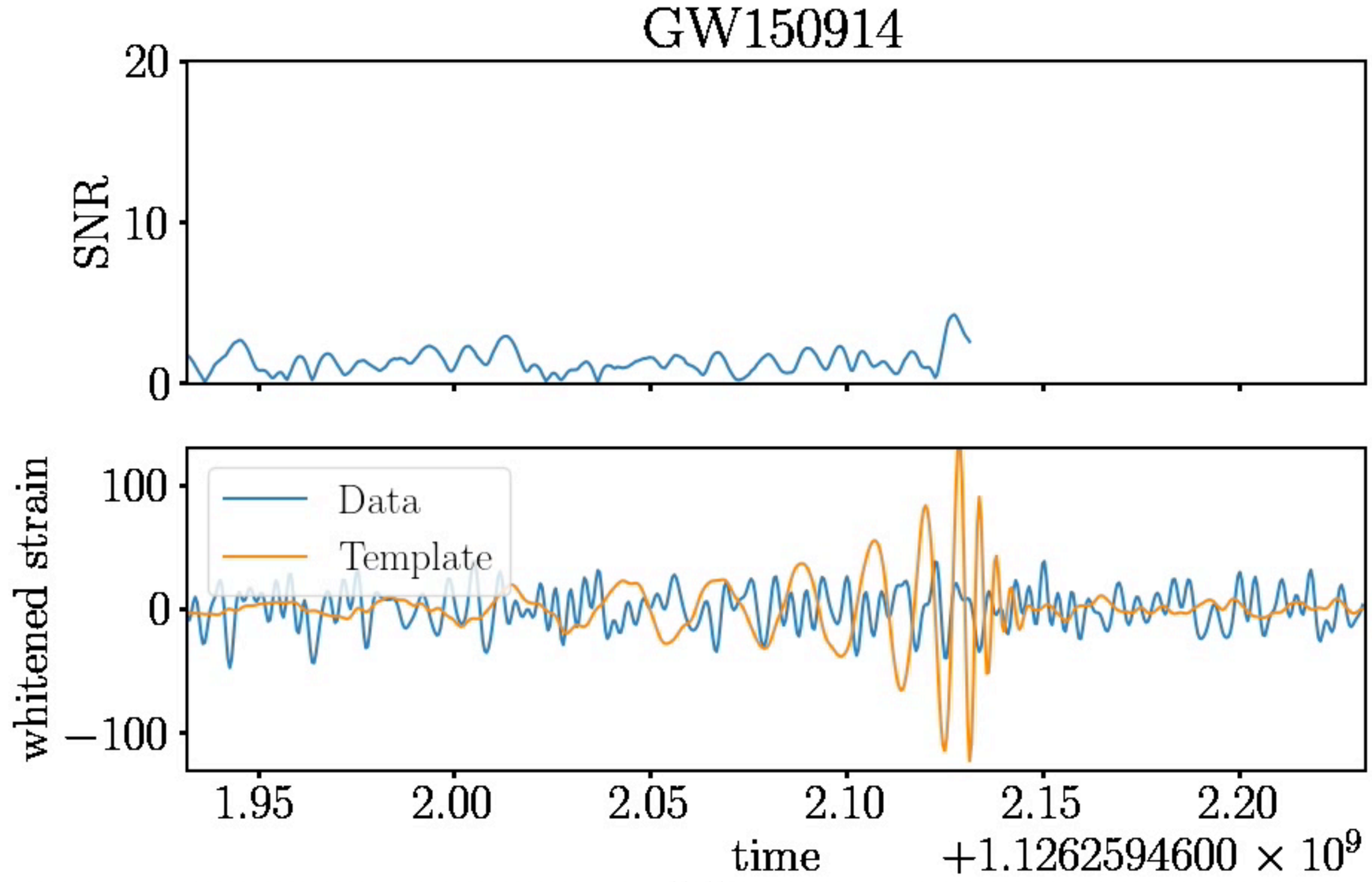
MAX-PLANCK-GESELLSCHAFT

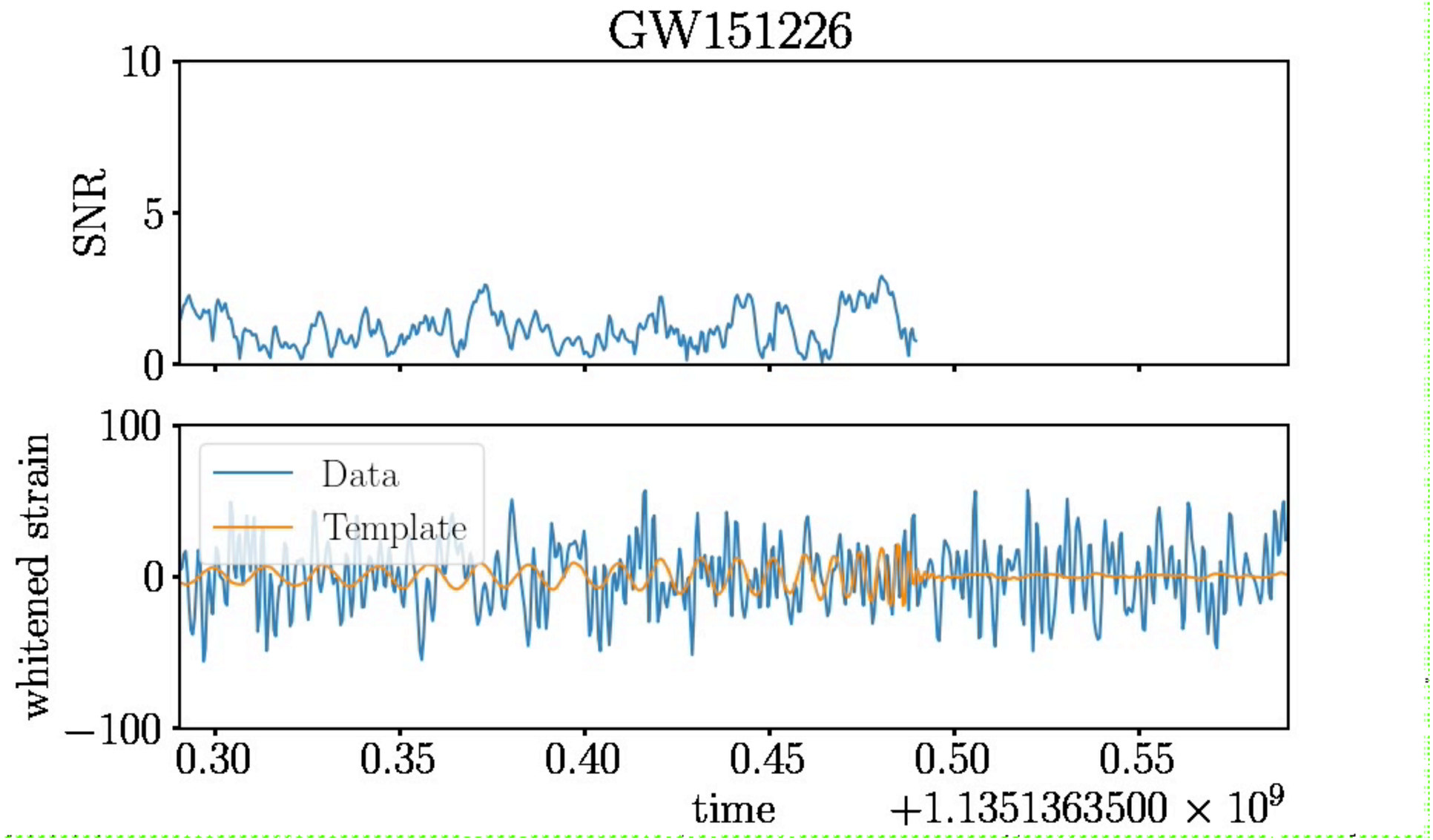
Matched Filtering

We can also interpret the matched filter as the **convolution** (filtering) of the **whitened data** $d_w(t)$ with the **whitened expected signal** $h_w(t)$ **:**

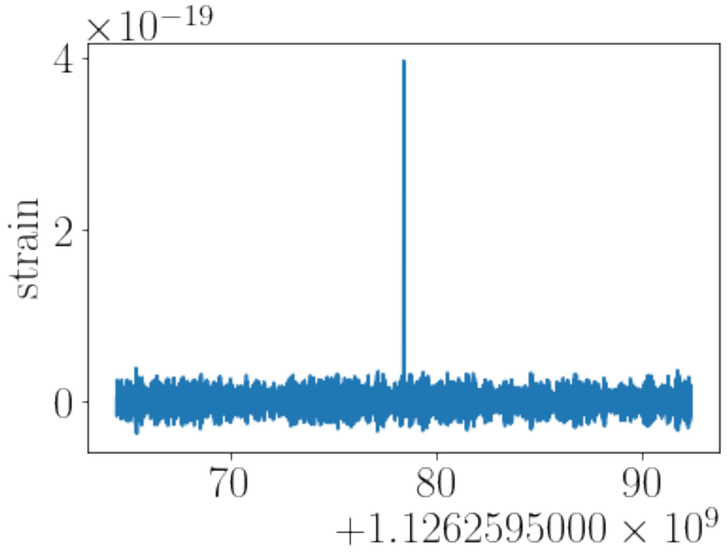
$$\hat{s} = (d\,|\,h) = 4\mathrm{Re}\int_0^\infty df \frac{\tilde{d}(f)\tilde{h}(f)^*}{S_n(f)} = \int_{-\infty}^{\infty} df \frac{\tilde{d}(f)}{\sqrt{S_n(f)/2}} \frac{\tilde{h}^*(f)}{\sqrt{S_n(f)/2}} = \int_{-\infty}^{\infty} dt d_w(t) h_w(t)$$

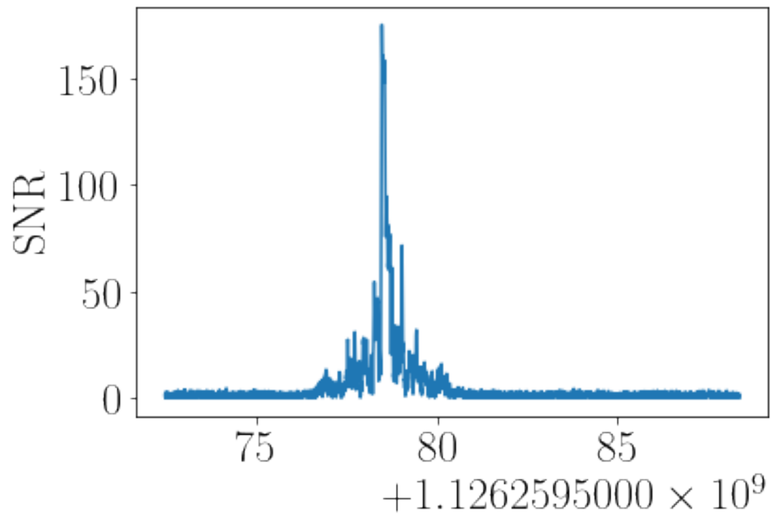



GW150914
SNR
20
10
0
whitened strain
100
0
−100
Data
Template
1.95
2.00
2.05
2.10
2.15
2.20
time
+1.1262594600 × 10^9


GW151226
SNR
10
5
0
whitened strain
100
0
−100
Data
Template
0.30
0.35
0.40
0.45
0.50
0.55
time
+1.1351363500 × 10^9

However, non-Gaussian glitches





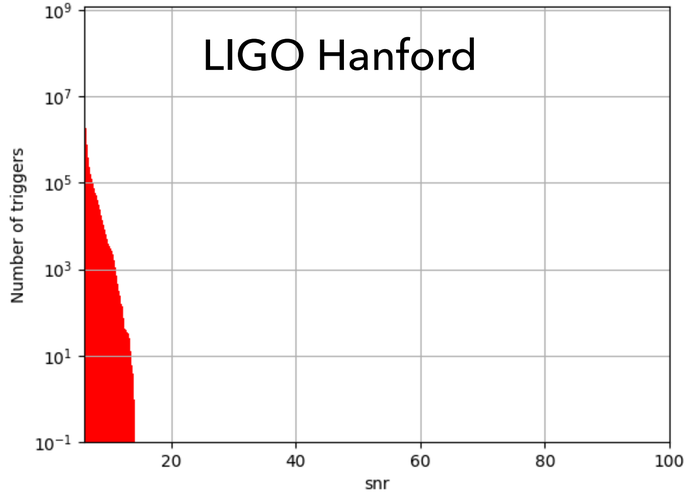
a = 4e-19

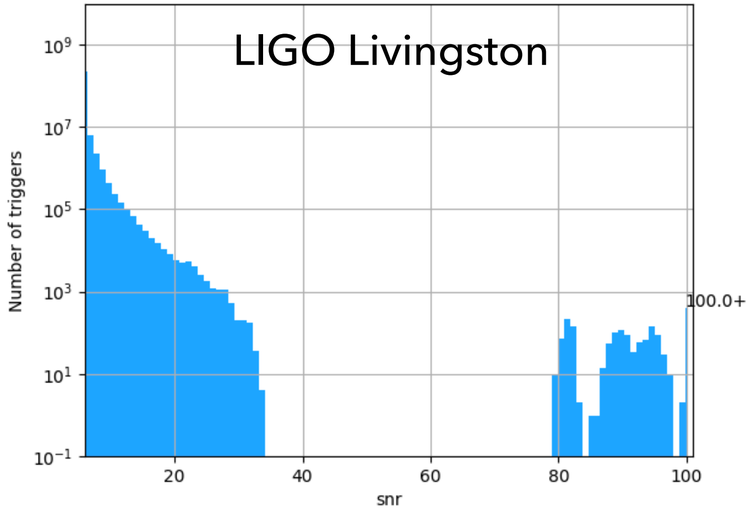
s = 0.01

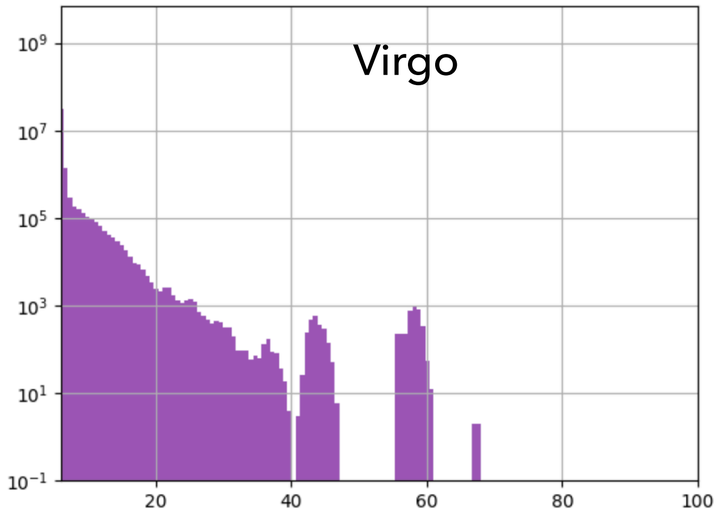
pulse = a*np.exp(-(t-t_0)**2/s**2)

但是，信噪比是一个好的Detection statistic吗?

Number of Events





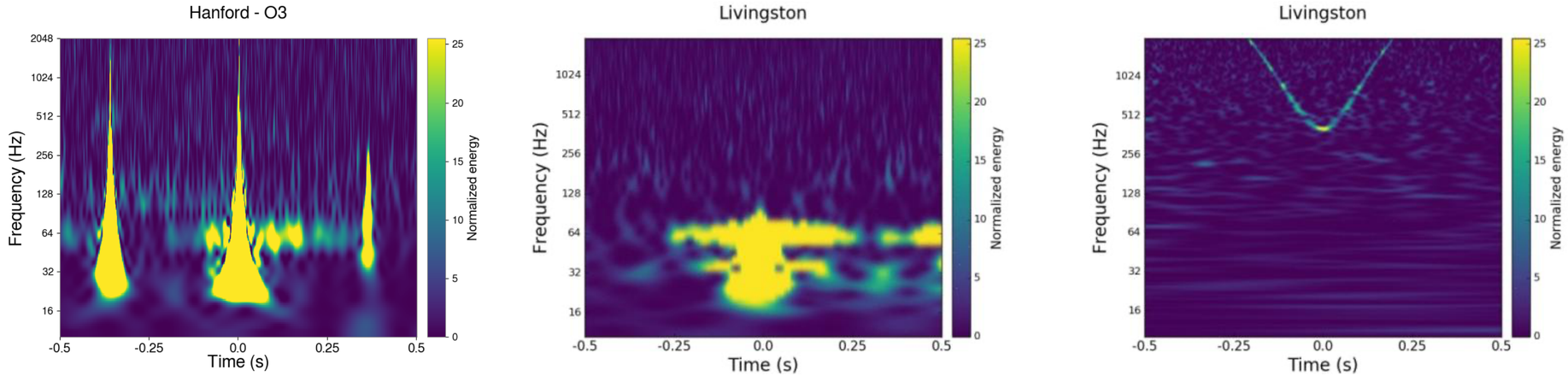


SNR

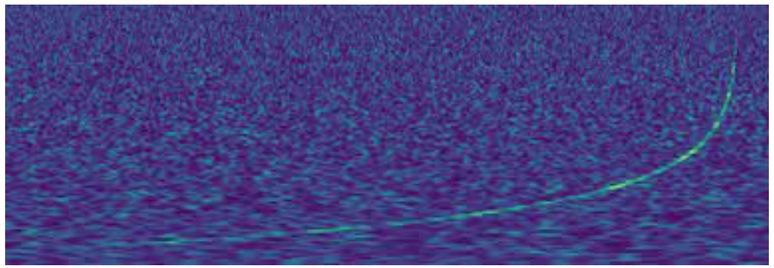
MAX-PLANCK-GESELLSCHAFT

非高斯噪音

- Glitches:

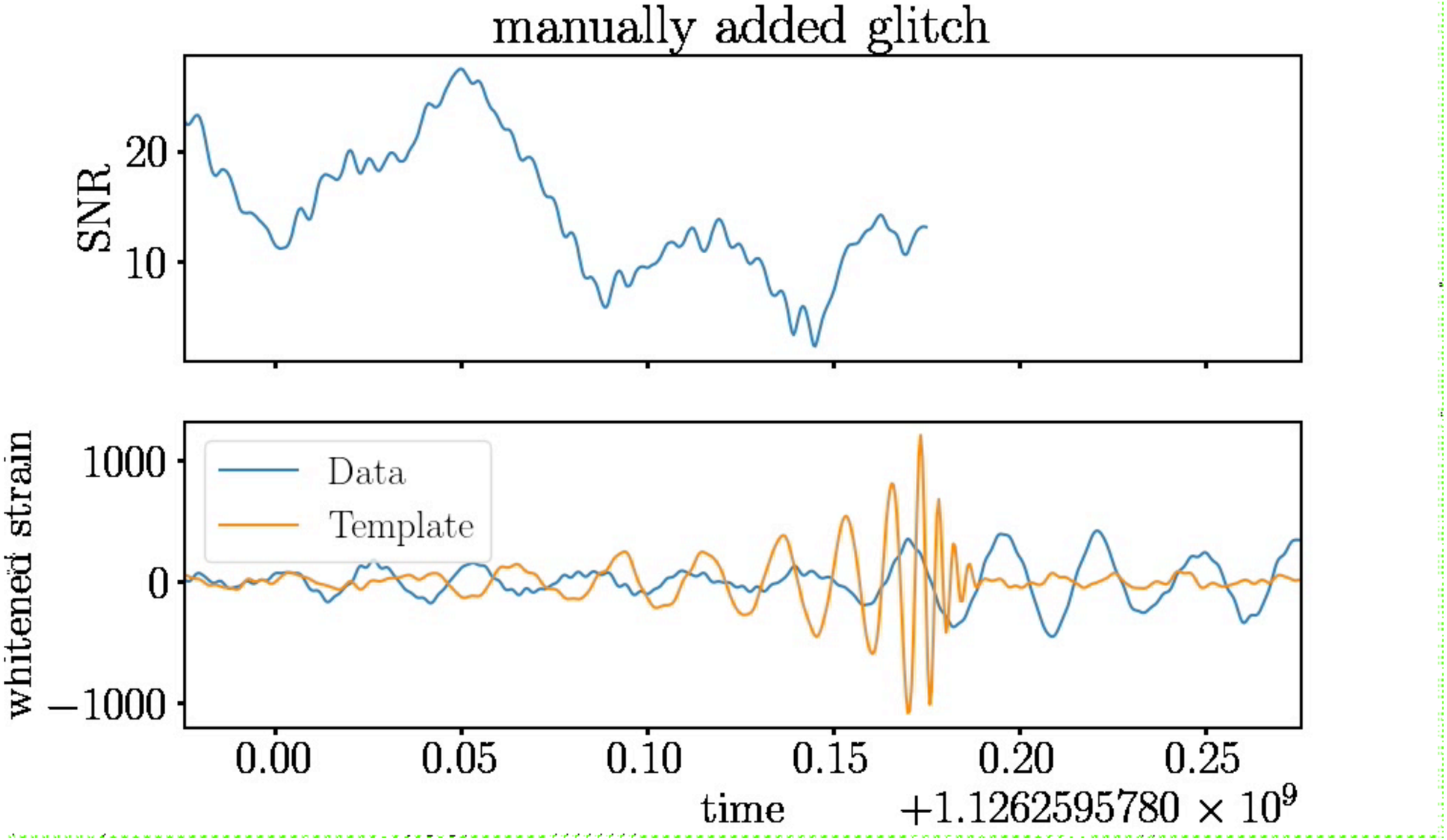


- Real Gravitational Waves:

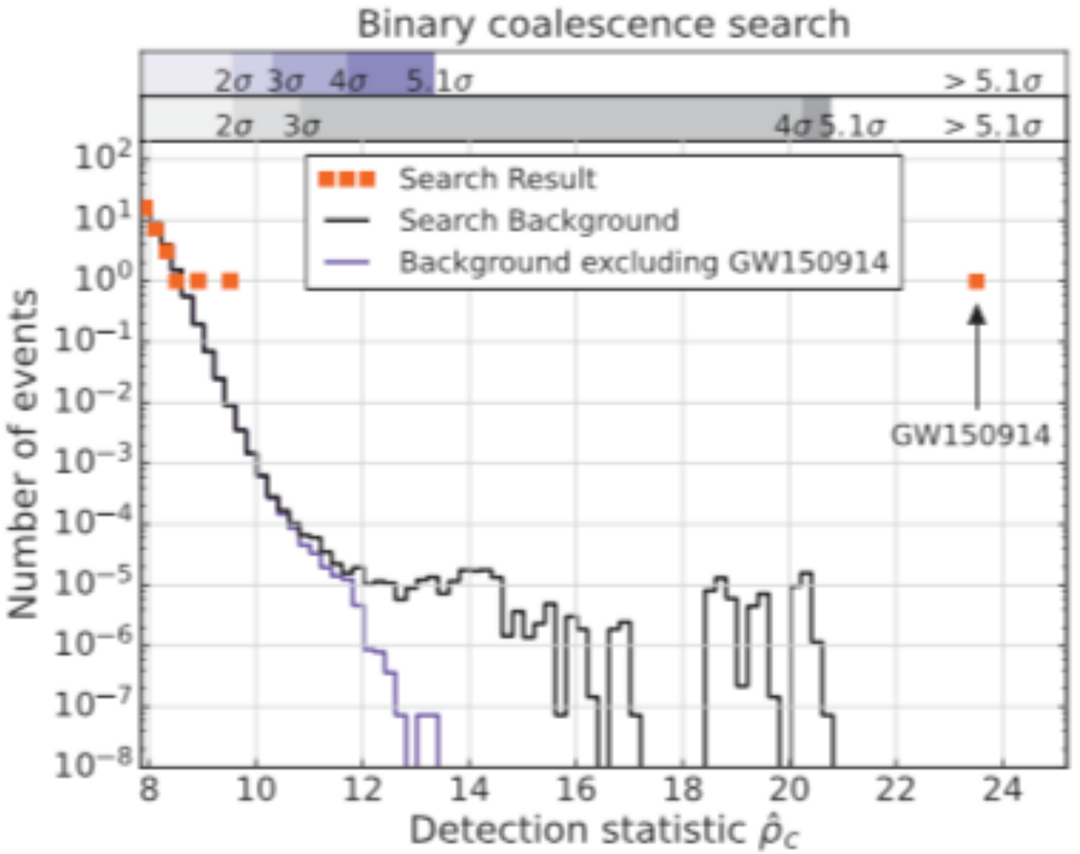


Time frequency trace of GW170817

MAX-PLANCK-GESELLSCHAFT


manually added glitch
SNR
20
10
whitened strain
1000
0
−1000
Data
Template
0.00
0.05
0.10
0.15
0.20
0.25
time
+1.1262595780 × 10^9

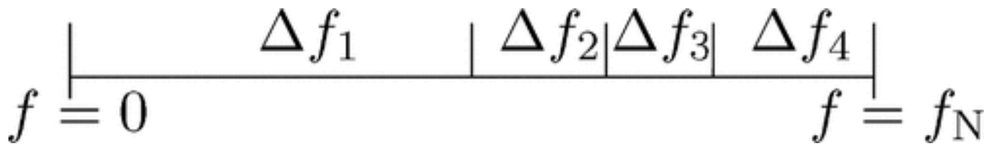
How to search CBC?



PRL 116, 061102 (2016)

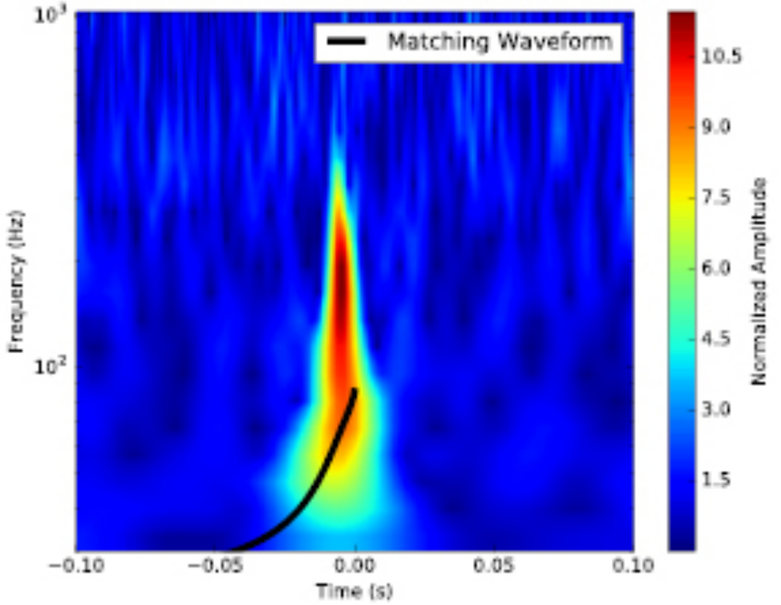
False alarm rate > 1/203000 year!

- 拟合优度检验：



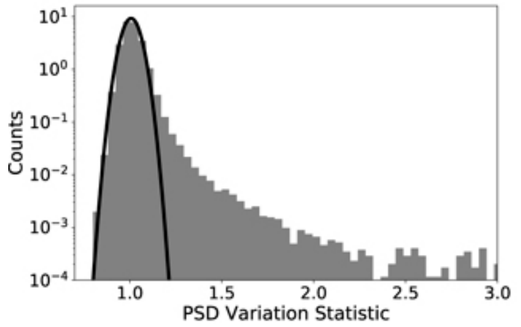
"Bruce Allen" chi**2 test, Phys. Rev. D **71**, 062001

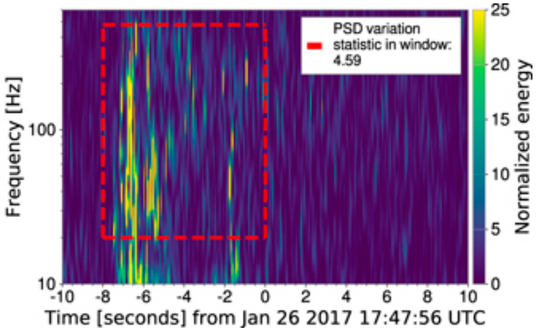
- Sine-Gaussian 检验：



Alex Nitz, sine Gaussian chi**2, Class. Quantum Grav. **35** 035016

- PSD方差偏离chi^2的检验：

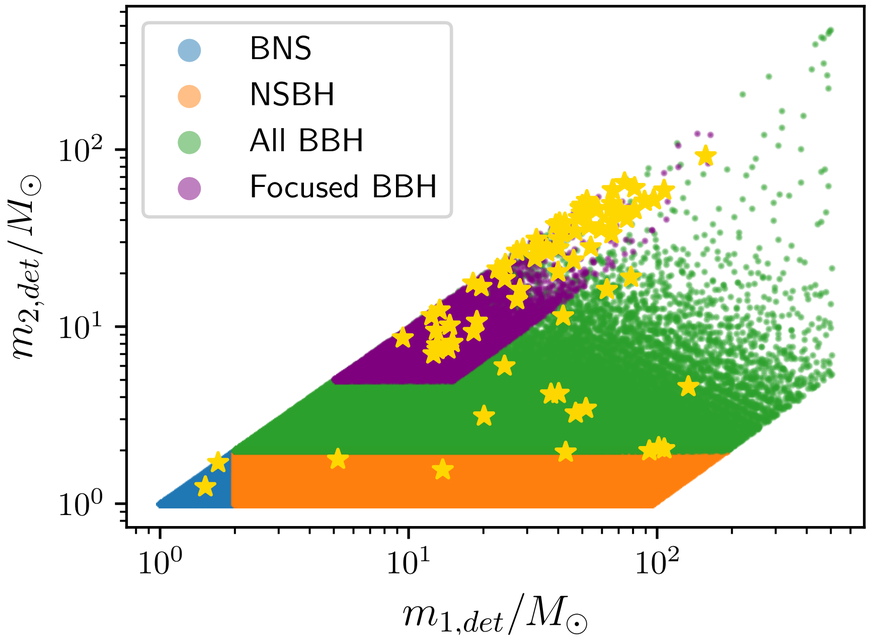




S Mozzon *et al* 2020 *Class. Quantum Grav.* **37** 215014

L(astrophysical origin) vs L(terristrial origin)

- 对噪音统计分布的更好理解 （L_noise）
- 对信号族群分布的更好理解（L_astro）



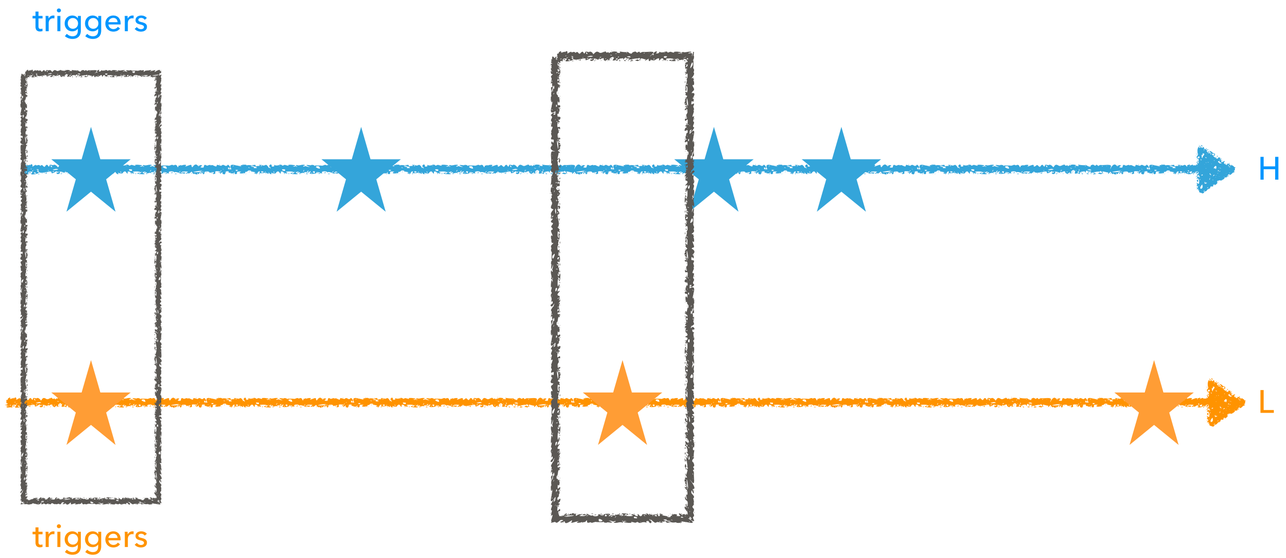
Made by Alex Nitz, 4-OGC paper

$$p_{\text{astro}} = \frac{L_{astro}}{L_{astro} + L_{noise}}$$

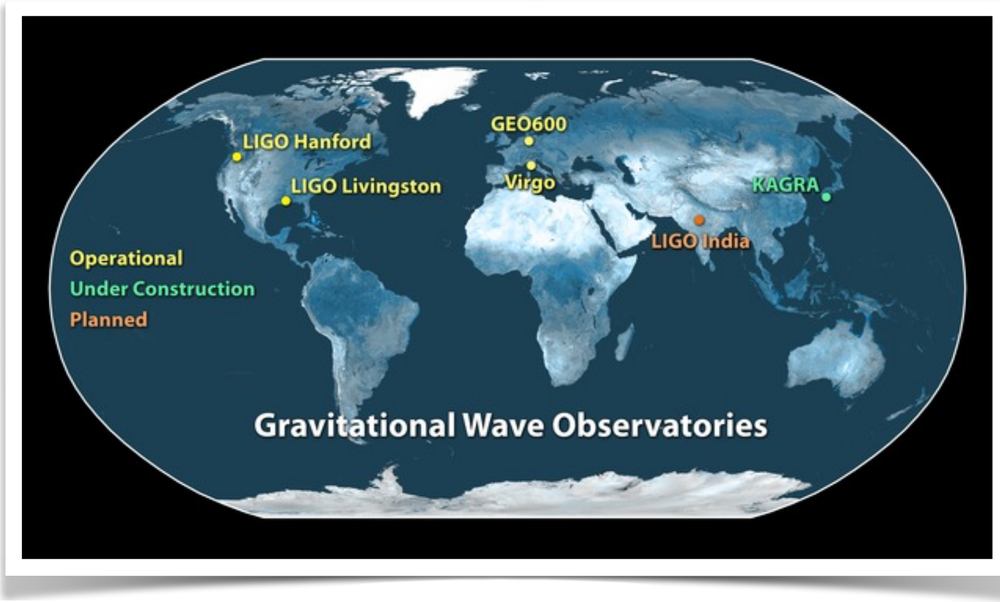
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Coincidence

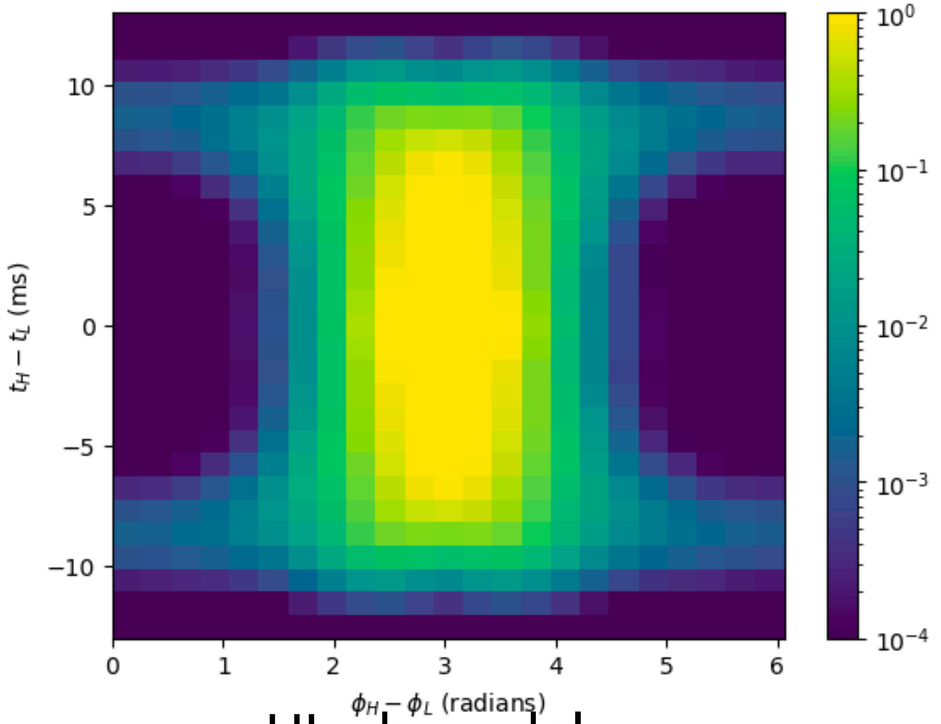
- By correlating triggers from multiple detectors to get false alarm rate (p value)



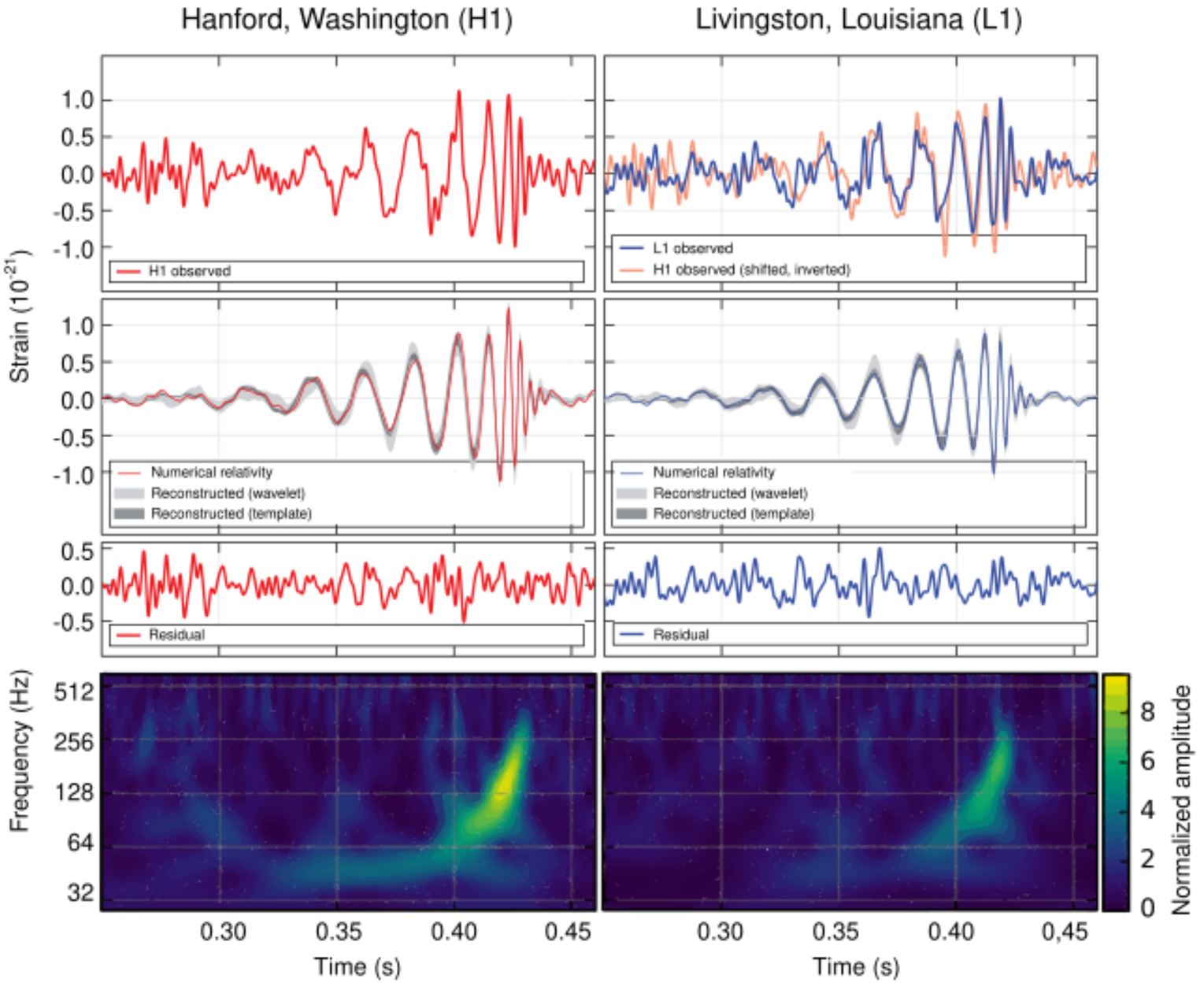
Time sliding the data to compute p value!



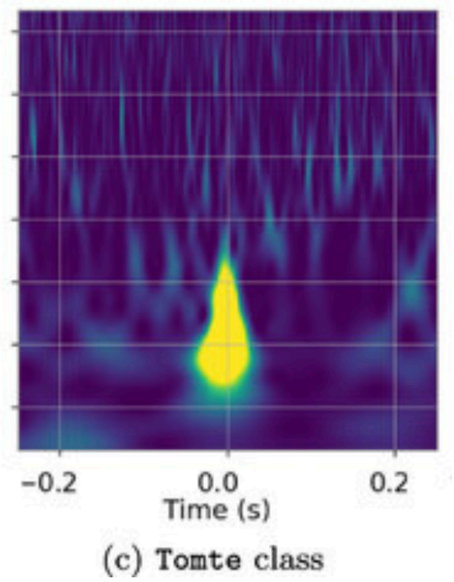
HL time delay



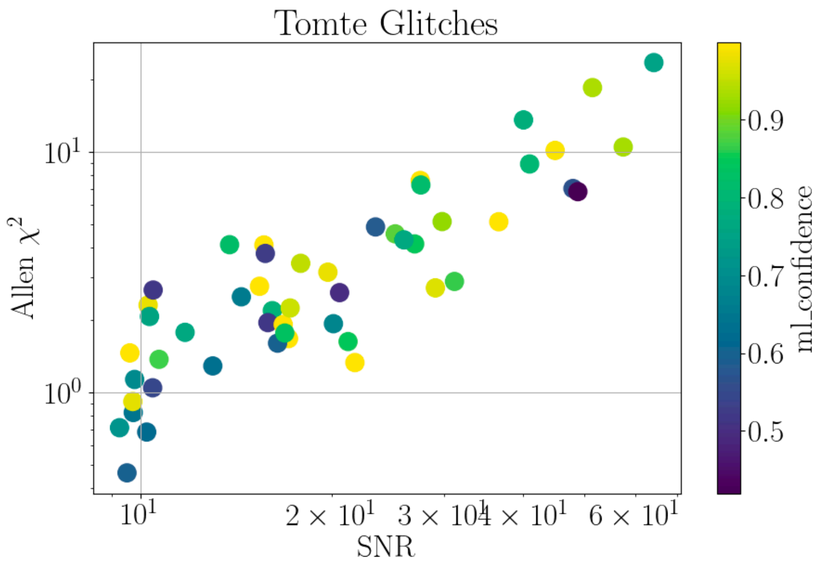
HL phase delay

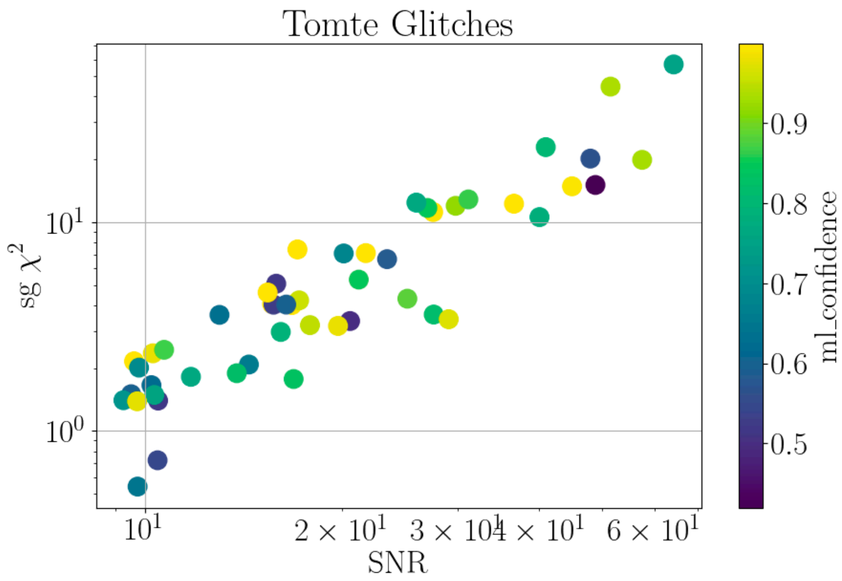

Hanford, Washington (H1)
Livingston, Louisiana (L1)
Strain (10^{-21})
1.0
0.5
0.0
-0.5
-1.0
H1 observed
L1 observed
H1 observed (shifted, inverted)
Numerical relativity
Reconstructed (wavelet)
Reconstructed (template)
Residual
Frequency (Hz)
512
256
128
64
32
0.30
0.35
0.40
0.45
0,45
Time (s)
Normalized amplitude
8
6
4
2
0

Trigger veto via other channels, e.g., gravityspy



(c) Tomte class





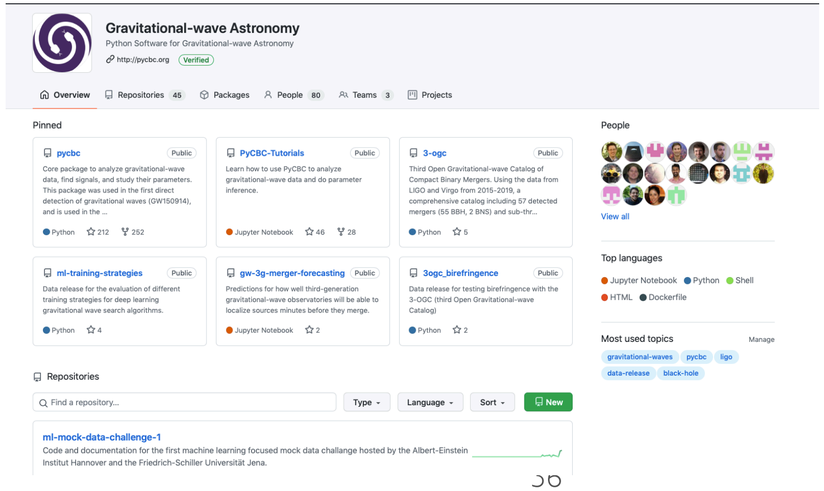
Tomte glitch: likely occurring while the bias voltage on a circuit (an electrostatic actuator) was adjusted.

Gravity Spy is an innovative citizen-science meets Machine Learning meets gravitational wave physics project.

Analysis of open data

- More Compact Binary Coalescence (CBC) events are yet to be discovered
  - AEI Hannover： OGC (Open Gravitational-wave Catalog）
- "Open" in the sense:

- Open data： www.gw-openscience.org
- Open-source software： www.pycbc.org
- Open, reproducible results： github.com/gwastro



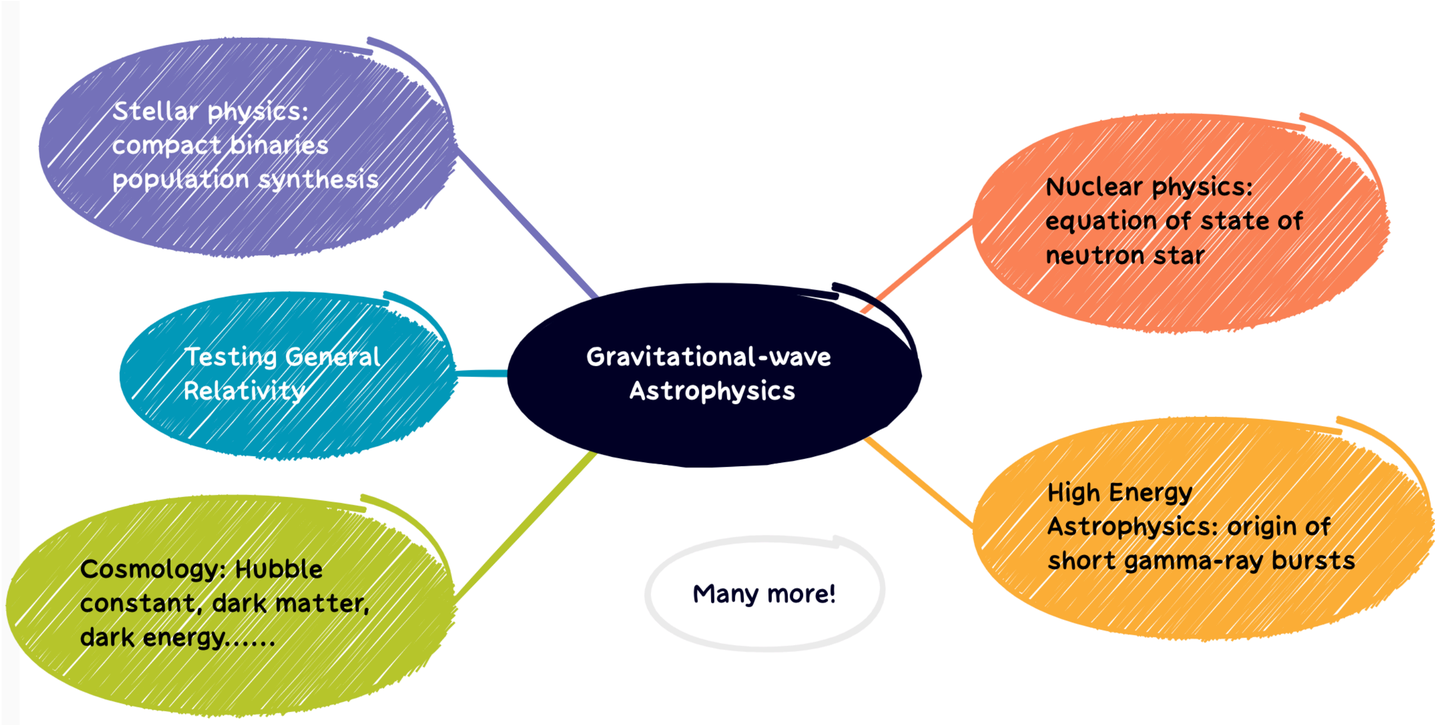
PyCBC

MAX-PLANCK-GESELLSCHAFT

SYLLABUS

- 引力波一般回顾
- 参数估计
- 黑洞族群分析/层级参数估计
- 引力波搜索

- 引力波是理论研究（广义相对论、黑洞） +
- 数值模拟（数值相对论） +
- 天文观测（引力波观测持、多信使观测续进行）+
- 数据分析（数据分析技术不断提升）+
- 物理应用的典范！



BACKUP SLIDES

Fisher Information Matrix

$p(\boldsymbol{\lambda}, d) = \dfrac{\mathscr{L}(d, \boldsymbol{\lambda})}{Z(d)}\pi(\boldsymbol{\lambda})$ Hard to compute, analytically or numerically (high dimensional parameter space)

Linear signal approximation: expand $\hat{h}(\boldsymbol{\lambda})$ linearly around **maximum likelihood estimator**

$$\hat{h}(\boldsymbol{\lambda}_{\mathrm{MLE}} + \Delta\boldsymbol{\lambda}) = \hat{h}(\boldsymbol{\lambda}_{\mathrm{MLE}}) + \sum_j \frac{\partial \hat{h}(\boldsymbol{\lambda})}{\partial \lambda_j}\Delta\lambda_j + \ldots \text{ with } \Delta\boldsymbol{\lambda} = \boldsymbol{\lambda} - \boldsymbol{\lambda}_{\mathrm{MLE}}$$

$$\begin{aligned}(d - \hat{h}(\boldsymbol{\lambda}) \,|\, d - \hat{h}(\boldsymbol{\lambda})) \approx (d - \hat{h}(\boldsymbol{\lambda}_{\mathrm{MLE}}) \,|\, d - \hat{h}(\boldsymbol{\lambda}_{\mathrm{MLE}})) \\ - \sum_j 2(d - \hat{h}(\boldsymbol{\lambda}_{\mathrm{MLE}}) \,|\, \partial_j \hat{h}(\boldsymbol{\lambda}))\Delta\lambda_j \\ + \sum_{ij} (\partial_i \hat{h}(\boldsymbol{\lambda}) \,|\, \partial_j \hat{h}(\boldsymbol{\lambda}))\Delta\lambda_i\Delta\lambda_j\end{aligned}$$

This is a constant that we can reabsorbe in the normalisation factor

Fisher Information Matrix

Assuming $d(t) - \hat{h}(t, \boldsymbol{\lambda}_{\mathrm{MLE}}) \approx n(t)$, the expected value of the second term is 0:

$$\langle (d - \hat{h}(\boldsymbol{\lambda}_{\mathrm{MLE}}) \,|\, \partial_j \hat{h}(\boldsymbol{\lambda})) \rangle_n = (\langle n \rangle \,|\, \partial_j \hat{h}(\boldsymbol{\lambda})) = 0$$

Defining the **Fisher matrix** $\Gamma_{ij} = (\partial_i \hat{h} \,|\, \partial_j \hat{h})$ and assuming **uniform priors:**

$$p(\boldsymbol{\lambda}, d)_{\mathrm{LSA}} \propto \mathscr{L}(d, \boldsymbol{\lambda})_{\mathrm{LSA}} \propto \exp\left[-\frac{1}{2} \Gamma_{ij} \Delta\lambda_i \Delta\lambda_j \right]$$

This is a **multivariate Gaussian distribution** for $\boldsymbol{\lambda}$ with **mean** $\boldsymbol{\mu} = \boldsymbol{\lambda}_{\mathrm{MSE}}$ and covariance matrix:

$$\Sigma_{ij} \equiv (\Gamma_{ij})^{-1}$$

Numerical Sampling, Monte Carlo

In the general case, we **cannot compute exactly the posterior distribution**, but we can numerically **sample** $\mathscr{L}(d, \boldsymbol{\lambda})\pi(\boldsymbol{\lambda})$ across **parameter space** to draw samples $\{\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \ldots, \boldsymbol{\lambda}_M\}$ from the posterior $p(\boldsymbol{\lambda}, d)$.

With **enough samples**, we can then approximate any integral involving the posterior:

$$\int f(\boldsymbol{\lambda}) p(\boldsymbol{\lambda}, d) d\boldsymbol{\lambda} \approx \frac{1}{M} \sum_i f(\boldsymbol{\lambda}_i)$$

and compute **marginal distributions**, **point estimates** and **credible intervals** from the samples.

Numerical Sampling, Monte Carlo

Typically **stochastic methods** are employed, since we cannot explore the high-dimensional parameter space with gridded approaches.

Two main techniques are commonly employed in GW astronomy: **Markov-Chain Monte-Carlo methods (MCMC) and Nested Sampling.**

The python parameter estimation software **Bilby** is the standard tool used by the LVK collaboration for estimating parameters from detected GW signals.



There are new emergent approaches that use **neural networks** to learn the posterior distributions, and employ **importance sampling** techniques to get accurate results: **DINGO** (developed by MPI-IS and this institute)

dingo-gw/**dingo**

Dingo: Deep inference for gravitational-wave observations

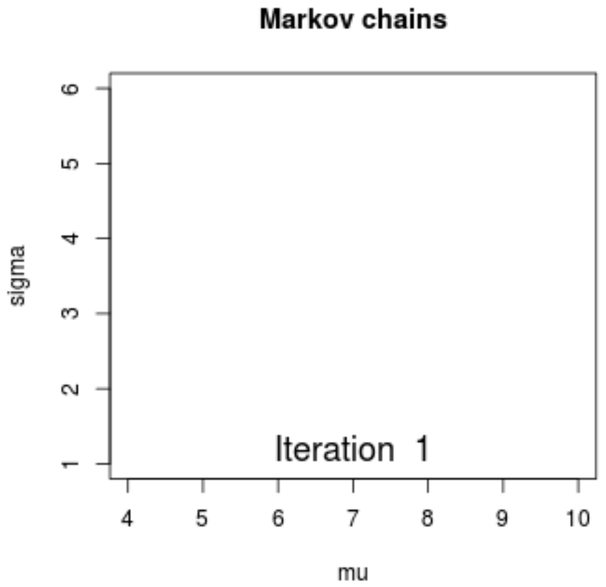
Numerical Sampling, Monte Carlo

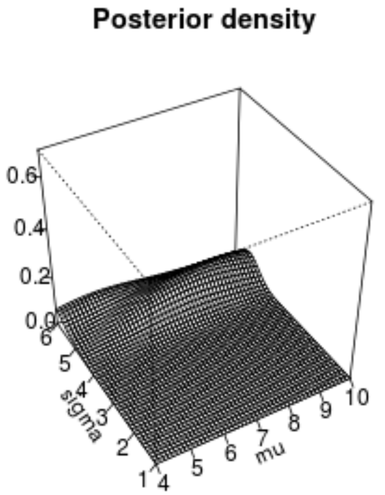
The idea is to generate samples using **Markov chains,** a type of random process where the probability of each event only depends on the previous event.

A **proposal distribution** is defined $Q(\lambda, \lambda')$ for generating a new sample λ' based on the last sample in the chain λ.

The new sample is accepted with probability min(1,α) where:

$$\alpha = \frac{Q(\lambda, \lambda')p(\lambda', d)}{Q(\lambda', \lambda)p(\lambda, d)}$$





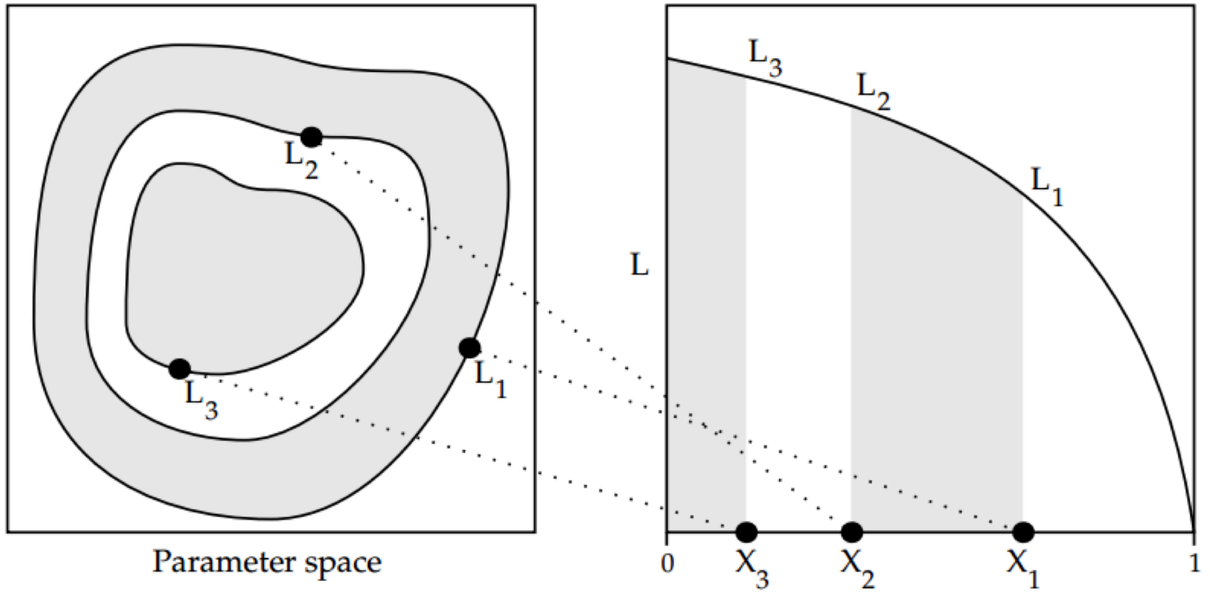
Numerical Sampling, Monte Carlo

Nested sampling is a technique for evaluating the **evidence** $Z(d)$ mapping the multidimensional problem into a 1D-problem (using **Lebesgue integration**):

$$Z(d) = \int d\boldsymbol{\lambda}\mathscr{L}(\boldsymbol{\lambda} \mid d)\pi(\boldsymbol{\lambda})$$

$$= \int_0^1 dX\mathscr{L}(X)$$

Where: $X(\mathscr{L}^*) = \int_{\mathscr{L}(\boldsymbol{\lambda}|d)>\mathscr{L}^*} d\boldsymbol{\lambda}\pi(\boldsymbol{\lambda})$



Evidence can be computed numerically:

$$Z = \sum_i^M \frac{1}{2}(X_{i-1} - X_{i+1}L_i)$$

OUTLINE

- Bayes' theorem: posterior = likelihood * prior / evidence
  - $P(\vec{\theta}\,|\,d,H) = \dfrac{\mathscr{L}(d\,|\,\vec{\theta},H)\pi(\theta\,|\,H)}{\mathscr{Z}(d\,|\,H)}$
- How to calculate the posterior?
  - Stochastic sampling: Markov Chain Monte Carlo (MCMC) / Nested Sampling
  - Downside: computationally expensive
- Accelerate the Bayes inference
  - New methods of sampling (e.g., non-Markov Chain)
  - Speeding up the calculation of likelihoods by using reduced representations
  - Parameter reduction by marginalization (e.g., analytical marginalization / numerical marginalization)
  - Machine learning
    - Neural network enhanced sampling (normalizing flow)
    - Directly produce the posterior (DINGO)